

IDENTIFICATION OF COVID FROM CXR
A MINI PROJECT REPORT

Submitted by

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Learn Beyond

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BONAFIDE CERTIFICATE

Certified that this project report titled “**Identification of Covid from CXR**” is the bonafide work of **Komal T (19CS074), Meenalakshmi S (19CS086), Mithra Devi R (19CS088)** who carried out the project work under my supervision. Certified further that to the best of my knowledge the work reported here in does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

The main aim of the project is to classifying X-ray images into three classes: normal, pneumonia, and COVID-19, based on two open-source datasets. Our data contains 180 X-ray images that belong to persons infected with COVID-19, and we attempted to apply methods to achieve the best possible results. In this project, we introduce some training techniques that help the network learn better when we have an unbalanced dataset (fewer cases of COVID-19 along with more cases from other classes). We also propose a neural network that is a concatenation of the Xception and ResNet50V2 networks. This network achieved the best accuracy by utilizing multiple features extracted by two robust networks. For evaluating our network, we have tested it on 11302 images to report the actual accuracy achievable in real circumstances. The average accuracy of the proposed network for detecting COVID-19 cases is 99.50%, and the overall average accuracy for all classes is 91.4%.

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CHAPTER 1

INTRODUCTION

1.1 Overview

The pervasive spread of the coronavirus around the world has quarantined many people and crippled many industries, which has had a devastating effect on human life quality, due to the high transmissibility of coronavirus, the detection of this disease (COVID-19) plays an important role in controlling it and planning preventative measures.

On the other hand, demographic conditions such as age and sex of individuals and many urban parameters such as temperature and humidity affect the prevalence of this disease in different parts of the world, which is more effective in spreading this disease

The lack of detective tools and the limitations in their production has slowed disease detection as a result, it increases the number of patients and casualties. The incidence of other diseases and the prevalence and number of casualties due to COVID-19 disease will decrease if it is detected quickly.

The first step is detection, recognize the symptoms of the disease, and use distinctive signs to detect the coronavirus accurately. Depending on the type of coronavirus, symptoms can range from those of the common cold to fever, cough, shortness of breath, and acute respiratory problems. The patient may also have a few days of cough for no apparent reason.

Unlike SARS, coronavirus affects not only the respiratory system but also other vital organs, such as the kidneys and liver. Symptoms of a new corona virus usually begin a few days after the person becomes infected, where, in some people, the symptoms may appear a little later. According to WHO, respiratory problems are one of the main symptoms of COVID-19, which can be detected by X-ray imaging of the chest. CT scans of the chest can also show the disease when symptoms are mild, so analyzing these images can detect the presence of the disease in suspicious people and even without initial symptoms.

Using these data can also overcome the limitations of other tools, such as lack of diagnostic kits and limitations of their production. The advantage of using CT scans and X-ray images is the availability of CT scan devices and x-ray imaging systems in most hospitals and laboratories, and the ease of access to the data needed to train the network and thus detect the disease. In the absence of common symptoms such as fever, the use of CT scans and X-ray images of the chest has a relatively good ability to detect the disease.

The use of specialists to diagnose the disease is a common method of detecting COVID-19 in laboratories. In this method, the specialist uses the symptoms and injuries in the chest radiology image to detect COVID-19 disease from a healthy person or person that is suffering from other diseases. This procedure has significant cost.

In recent years, computer vision and Deep Learning have been used to detect many different diseases and lesions in the body automatically. Some examples are: Detection of tumor types and volume in lungs, breast, head and brain state-of-the-art bone suppression in x-rays, diabetic retinopathy classification, prostate segmentation, nodule classification; skin lesion classification, analysis of the myocardium in coronary CT angiography; sperm detection and tracking etc.

CT scan or X-ray images analysis is one of the methods of diagnosing COVID-19, the use of computer vision and Deep Learning can play a beneficial role in diagnosing this disease. Due to the sensitivity of the Covid-19 diagnosis. On the other hand, our focus is on increasing the detection efficiency due to the limited open-source data available.

Due to the sensitivity of the Covid-19 diagnosis, the diagnostic accuracy is one of the main challenge faced. On the other hand, our focus is on increasing the detection efficiency due to the limited open-source data available.

To improve COVID-19 detection and diminish false COVID-19 detections. This is done by combining two robust deep convolutional neural networks and optimizing the training parameters. Besides, we also propose a method for training the network when the dataset is imbalanced.

In statistical analysis of CT scans was performed by several specialists and diagnosticians, who classified the suspects into several classes for diagnosis and treatment. Because of the superiority of computer vision and Deep Learning in the analysis of medical images, after the reliability of CT scans of the chest for COVID-19 detection, artificial intelligence became useful to detect the disease and measure the rate of infection and damage to the lungs using CT scans and the course of the disease, with promising results.

On the other hand CT scans which produce ionizing radiation. Research shows that CT scans use X-rays, which are a type of radiation called ionizing radiation. It can damage the DNA in your cells and raise the chance that they'll turn cancerous. These scans expose you to more radiation than other imaging tests, like X-rays and mammograms. But radiation's effect adds up over your lifetime. So risk increases with every CT scan you get. So the main aim of the project is to detect COVID 19, pneumonia using chest X ray.

CHAPTER 2

REQUIREMENTS

2.1 SOFTWARE REQUIREMENTS

2.1.1 GOOGLE COLABORATORY

Colab is a free Jupyter notebook environment that runs entirely in the cloud. It does not require a setup and the notebooks that you create can be simultaneously edited by your team members. Colab supports many popular machine learning libraries which can be easily loaded in notebook.

2.1.2 PYTHON

Python is an interpreted high-level general-purpose programming language. Its design philosophy emphasizes code readability with its use of significant indentation. Its language constructs as well as its object-oriented approach aim to help programmers write clear, logical code for small and large-scale projects. Python is dynamically-typed and garbage-collected.

It supports multiple programming paradigms, including structured (particularly, procedural), object-oriented and functional programming. It is often described as a "batteries included" language due to its comprehensive standard library.

2.1.3 TENSERFLOW

TensorFlow offers multiple levels of abstraction so you can choose the right one for your needs. Build and train models by using the high-level Keras API, which makes getting started with TensorFlow and machine learning easy.

If you need more flexibility, eager execution allows for immediate iteration and intuitive debugging. For large ML training tasks, use the Distribution Strategy API for distributed training on different hardware configurations without changing the model definition.

TensorFlow has always provided a direct path to production. Whether it's on servers, edge devices, or the web, TensorFlow lets you train and deploy your model easily, no matter what language

or platform you use. Use TensorFlow Extended (TFX) if you need a full production ML .

2.1.4 KERAS

Keras is an API designed for human beings, not machines. Keras follows best practices for reducing cognitive load: it offers consistent & simple APIs, it minimizes the number of user actions required for common use cases, and it provides clear & actionable error messages.

Keras is the high-level API of TensorFlow 2: an approachable, highly-productive interface for solving machine learning problems, with a focus on modern deep learning. It provides essential abstractions and building blocks for developing and shipping machine learning solutions with high iteration velocity. Keras empowers engineers and researchers to take full advantage of the scalability and cross-platform capabilities of TensorFlow 2: you can run Keras on TPU or on large clusters of GPUs, and you can export your Keras models to run in the browser or on a mobile device.

2.1.5 OPEN CV

OpenCV (Open Source Computer Vision Library) is an open source computer vision and machine learning software library. OpenCV was built to provide a common infrastructure for computer vision applications and to accelerate the use of machine perception in the commercial products. Being a BSD-licensed product, OpenCV makes it easy for businesses to utilize and modify the code.

The library has more than 2500 optimized algorithms, which includes a comprehensive set of both classic and state-of-the-art computer vision and machine learning algorithms. These algorithms can be used to detect and recognize faces, identify objects, classify human actions in videos, track camera movements, track moving objects, extract 3D models of objects, produce 3D point clouds from stereo cameras, stitch images together to produce a high resolution image of an entire scene, find similar images from an image database, remove red eyes from images taken using flash, follow eye movements, recognize scenery and establish markers to overlay it with augmented reality, etc.

OpenCV has more than 47 thousand people of user community and estimated number of downloads exceeding 18 million. The library is used extensively in companies, research groups and by governmental bodies.

2.1.6 PYDICOM

pydicom is a pure python package for working with DICOM files such as medical images, reports, and radiotherapy objects. It is designed to let you manipulate data elements in DICOM files with python code. pydicom is easy to install and use, and because it is a pure python package, it should run anywhere python runs.

pydicom is not a DICOM server, and is not primarily about viewing images. It is designed to let you manipulate data elements in DICOM files with python code. pydicom is easy to install and use, and because it is a pure python package, it should run anywhere python runs. One limitation of pydicom: compressed pixel data (e.g. JPEG) cannot be altered in an intelligent way as it can be for uncompressed pixels. Files can always be read and saved, but compressed pixel data cannot easily be modified.

2.1.7 PANDAS

Pandas is a Python package providing fast, flexible, and expressive data structures designed to make working with “relational” or “labeled” data both easy and intuitive. It aims to be the fundamental high-level building block for doing practical, real-world data analysis in Python. Additionally, it has the broader goal of becoming the most powerful and flexible opensource data analysis/manipulation tool available in any language. It is already well on its way toward this goal.

2.1.8 NUMPY

NumPy, which stands for Numerical Python, is a library consisting of multidimensional array objects and a collection of routines for processing those arrays. Using NumPy, mathematical and logical operations on arrays can be performed.

NumPy is a Python package. It stands for ‘Numerical Python’. It is a library consisting of multidimensional array objects and a collection of routines for processing of array.

2.2 FUNCTIONAL REQUIRMENTS

2.2.1 PROPER DETECTION ON UNBALANCED DATASETS

In this project, we have introduced a training method in which the network learns better even the covid-19 cases are less in number. Since there are less number of covid-19 Chest X Ray images available, we have developed a model which learns and produce a efficent model which delivers good results when the dataset is unbalanced.

2.3 NON-FUNCTIONAL REQUIRMENTS

2.3.1 RELIABILITY

This software will be developed with machine learning, feature engineering and deep learning techniques. So, in this step there is no certain reliable percentage that is measurable. Also, user provided data will be used to compare with result and measure reliability. With recent machine learning techniques, user gained data should be enough for reliability if enough data is obtained.

2.3.2 ACCURACY

Accuracy is one of the most important metrics to analyze a deep learning model. The system should able to categorize the prvided image into normal or pnemonia or covid-19.

2.3.3 CAPACITY

The model can accurately detect irrespective of the size of the dataset

2.3.4 SUPPORTABILITY

The system should require C, Java, Python and Matlab knowledge to maintenance. If any problem acquire in server side and deep learning methods, it requires code knowledge and deep learning background to solve. Client side problems should be fixed with an update and it also require code knowledge and network knowledge

CHAPTER 3

MODULES DESCRIPTION

3.1 DATASET PREPARING

Two open-source datasets is in the project. The covid chestxray dataset is taken from this GitHub repository (<https://github.com/ieee8023/covid-chestxray-dataset>), This dataset consists of X-ray images of patients infected to COVID-19, SARS, Streptococcus, ARDS, Pneumocystis, and other types of pneumonia from different patients. In this dataset, we only considered the X-ray images, and in total, there were 180 images from 118 cases with COVID-19 and 42 images from 25 cases with Pneumocystis, Streptococcus, and SARS that were considered as pneumonia. The second dataset was taken from (<https://www.kaggle.com/c/rsna-pneumonia-detection-challenge>), which contains 6012 cases with pneumonia and 8851 normal cases.

The number of allocated images to training and validation set in both dataset

Dataset	COVID 19	Pneumonia	Normal
Covid chestxray dataset	180	42	0
rsna pneumonia detection challenge	0	6012	8851
Total	180	6054	8851
Training set	149	1634	2000
Validation set	41	4420	6851

Only 180 cases infected with COVID-19, which is few data for a class as compared to other classes. If many images from normal or pneumonia classes with few COVID-19 images combined for training, the network would become able to detect pneumonia and normal classes very well, but not the COVID-19 class because of the unbalanced dataset. In that case, although the network cannot identify

COVID-19 properly, as there are many more images of pneumonia and normal classes than the COVID-19 class, the general accuracy would become very high, but not the COVID-19 detection accuracy. This condition is not our goal because the main purpose here is to achieve good results in detecting COVID-19 cases and not to identify wrong COVID-19 cases.

The best way to solve this problem is to make the dataset balanced and provide the network almost equal data of each class when training, so that the network will learn to identify all classes. Here because not able access more open-source datasets of COVID-19 to increase this class data, we chose the number of pneumonia and normal classes almost equal to the COVID-19 number of images.

And decided to train the networks for 8 consecutive phases. In each phase, selected 250 cases of normal class and 234 cases of pneumonia class along with the 149 COVID-19 cases. In total, we had 633 cases for each training phase. All of the COVID-19 images and 34 of the pneumonia images were common between each training phase and 250 normal cases, and 200 pneumonia cases were unique in each training phase. The common 149 COVID-19 and 34 pneumonia cases between all the training phases were from the covid chestxray dataset; and the rest of the data were from the other dataset. Based on this categorizing, our training set includes 8 phases and 3783 images.

By doing this network sees an almost equal number of images for each class, so it helps to improve the COVID-19 detection along with detecting pneumonia and normal cases. But as we had more pneumonia and normal cases, we showed the network different pneumonia and normal cases with COVID-19 cases in each phase. Implementing this method results in two advantages. One is that the network learns COVID-19 class features better along with the other classes; second, the normal and pneumonia classes' detection improves greatly. Better detecting pneumonia and normal cases means not detecting wrong COVID-19 cases, which is one of our goals. Running this method also helps the network better identify COVID-19 and not detect faulty COVID-19 cases. This method can be used for all circumstances in which there is a highly unbalanced dataset.

3.2 TRAINING PHASE

In the training phase, we allocated 8 phases for training. For reporting more reliable results, we chose five folds for training, where in every fold the training set was made of 8 phases as it is mentioned. We have trained ResNet50V2; Xception; and a concatenation of Xception and ResNet50V2 neural networks based on the explained method. This concatenated Neural Network has shown higher accuracy compared to others. As we have tested several networks in our project, the Xception and ResNet50V2 networks work as well or better than others in extracting deep features. By concatenating the output features of both networks, we helped the network learn to classify the input image from both feature vectors, which resulted in better accuracy.

Parameter and function used in training procedure

Training Parameters	Xception	ResNet50V2	Concatenated
Learning Rate	1e-4	1e-4	1e-4
Batch Size	30	30	20
Optimizer	Nadam	Nadam	Nadam
	Categorical	Categorical	Categorical
Loss Function	Crossentropy	Crossentropy	Crossentropy
Epochs per each		100	100
Training Phase	100		
Horizontal/Vertical flipping	Yes	Yes	Yes
Zoom Range	5%	5%	5%
Rotation Range	0–360°	0–360°	0–360°
Width/Height	5%	5%	5%
Shift Range	5%	5%	5%
Re-scaling	1/255	1/255	1/255

Based on Table trained the networks using the Categorical cross-entropy loss function and

Nadam optimizer. The learning rate was set to $1e-4$. We trained the network for 100 epochs in each training phase and, because of having 8 training phases, the models were trained for 800 epochs. For the Xception and ResNet50V2, we selected the batch size equal to 30. But as the concatenated network had more parameters than Xception and ResNet50V2, we set the batch size equal to 20. Data augmentation methods were also implemented to increase training efficiency and prevent the model from overfitting.

3.3 DATASET VALIDATION

Validated our networks on 31 cases of COVID-19, 4420 cases of pneumonia, and 6851 normal cases. The reason our training data was less than the validation data is that we had a few cases of COVID-19 among many normal and pneumonia cases. Therefore, not able use many images from the two other classes with COVID-19 fewer cases for training, because it would have made the network not learn COVID-19 features.

To solve this issue, we selected 3783 images for training in 8 different phases. valuated our network on the remainder of the data so that our trained network's ultimate performance would be clear. It must be noticed that exceptionally, in fold3, we had 30 cases of COVID-19 for validation, and 150 other cases were allocated for training.

CHAPTER 4

DESIGN OF PROPOSED SYSTEM

4.1 ARCHITECTURE

Deep convolutional neural networks are useful in machine vision tasks. These have created advances in many field like Agriculture; medical disease diagnosis; and industry. The superiority of these networks comes from the robust and valuable semantic features they generate from input data. The main focus of deep networks is detecting infection in X-ray images, so classifying the X-ray images into normal, pneumonia or COVID-19. Some of the powerful and most used deep convolutional networks are VGG; ResNet; DenseNet ; Inception; Xception.

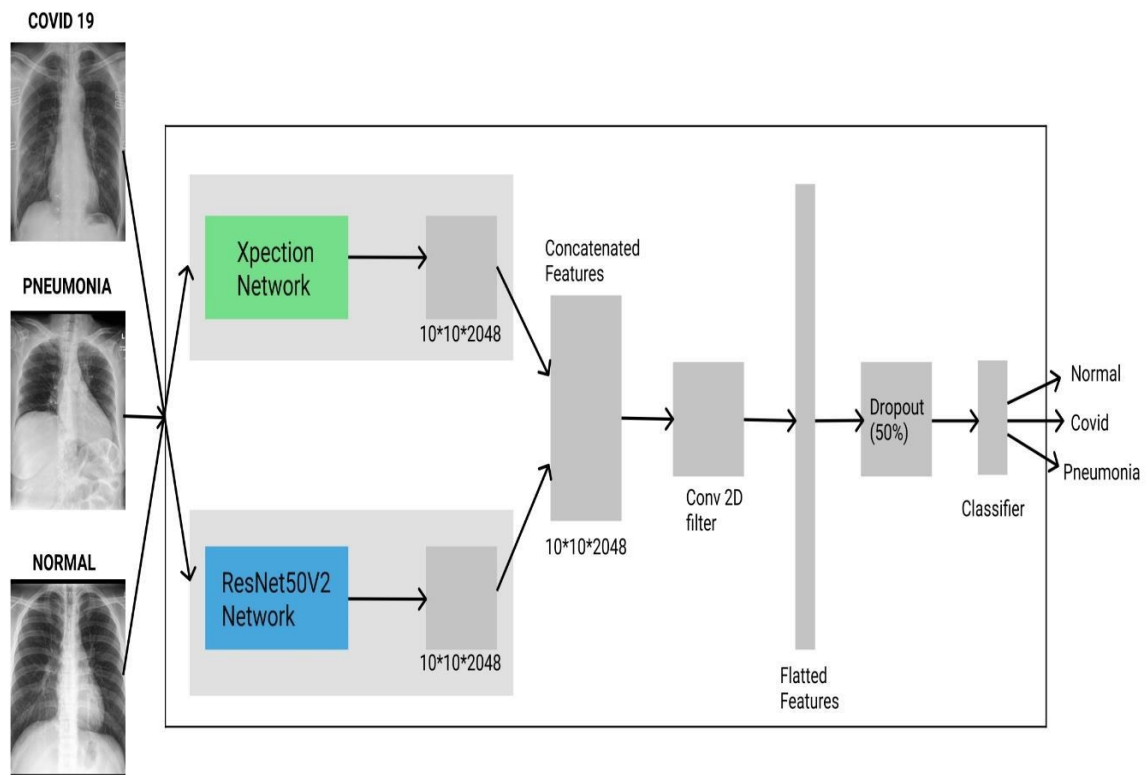


Fig 4.1 Architecture diagram

Xception is a deep convolutional neural network that introduced new inception layers. These inception layers are constructed from depthwise convolution layers, followed by a point-wise convolution layer. Xception achieved the third-best results on the ImageNet dataset after InceptionresnetV2 and NasNet Large.

ResNet50V2 is a modified version of ResNet50 that performs better than ResNet50 and ResNet101 on the ImageNet dataset. In ResNet50V2, a modification was made in the propagation formulation of the connections between blocks. ResNet50V2 also achieves a good result on the ImageNet dataset.

The pre-processed input images of our dataset are 300×300 pixels. Xception generates a $10 \times 10 \times 2048$ feature map on its last feature extractor layer from the input image, and ResNet50V2 also produces the same size of feature map on its final layer. As both networks generate the same size of feature maps, we concatenated their features so that by using both of the inception-based layers and residual-based layers, the quality of the generated semantic features would be enhanced.

A concatenated neural network is designed by concatenating the extracted features of Xception and ResNet50V2 and then connecting the concatenated features to a convolutional layer that is connected to the classifier that automatically orders or categorizes data into one or more of a set of “classes.”

The kernel size of the convolutional layer that was added after the concatenated features was 1×1 with 1024 filters and no activation function. This layer was added to extract a more valuable semantic feature out of the features of a spatial point between all channels, with each channel being a feature map. This convolutional layer helps the network learn better from the concatenated features extracted from Xception and ResNet50V2.

4.2 FLOWCHART

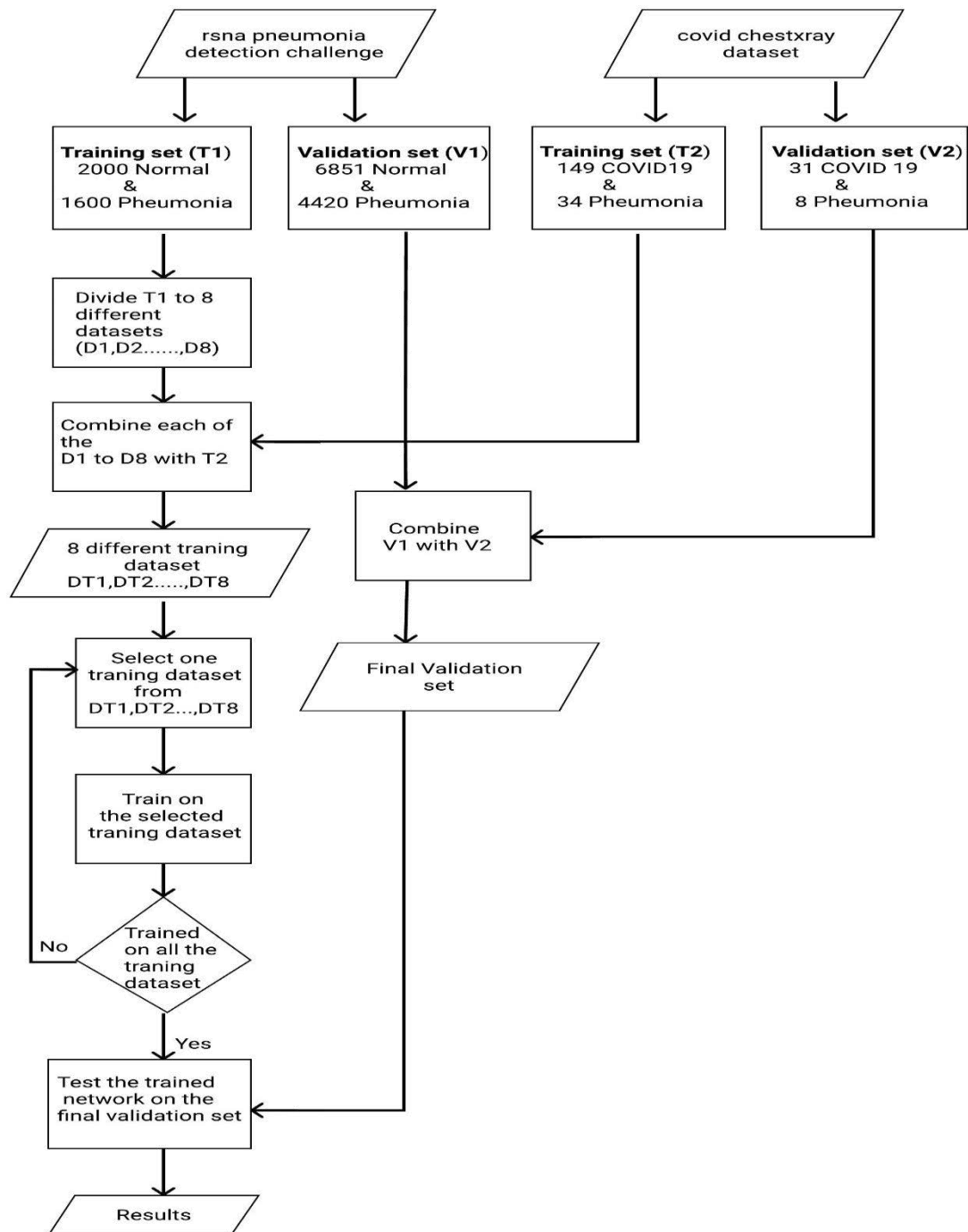


Fig 4.2 Flowchart

The two dataset rsna pneumonia and covid chestxray are divided into Training set (T1,T2) and Validation set (V1,V2). Training set (T1) consist of 2000 Normal and 1600 pneumonia case Validation set (V2) contain 6851 Normal and 4420 pneumonia cases Training set (T2) consist of 149 COVID 19 and 34 pneumonia cases Validation set (V2) contain 31 COVID 19 and 8 pneumonia cases.

Training set (T1) divided into different datasets D1 to D8 and combine each of D1 to D8 with Training set (T2) to get 8 different training dataset DT1 to DT8 this process is called data augmentation. Select one Training dataset from DT1 to DT8 which is tested with final validation set which is formed combining Validation set (V1,V2).

CHAPTER 5

IMPLEMENTATION OF PROPOSED SYSTEM

5.1 NEURAL NETWORKS

Deep convolutional neural networks are useful in machine vision tasks. These have created advances in many field like Agriculture; medical disease diagnosis; and industry The superiority of these networks comes from the robust and valuable semantic features they generate from input data. Here the main focus of deep networks is detecting infection in X-ray images, so classifying the X-ray images into normal, pneumonia or COVID-19. Some of the powerful and most used deep convolutional networks are VGG; ResNet; DenseNet; Inception; Xception.

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5.2 DATASET

Two open-source datasets in our work. The covid chestxray dataset is taken from this GitHub repository. This dataset consists of X-ray and CT scan images of patients infected to COVID-19, SARS, Streptococcus, ARDS, Pneumocystis, and other types of pneumonia from different patients. In this dataset, the X-ray images, and in total, there were 180 images from 118 cases with COVID-19 and 42 images from 25 cases with Pneumocystis, Streptococcus, and SARS that were considered as pneumonia. The second dataset was taken from [kaggle](#), it's contains 6012 cases with pneumonia and 8851 normal cases.

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As stated, only 180 cases infected with COVID-19, which is few data for a class as compared to other classes. If we had combined lots many images from normal or pneumonia classes with few COVID-19 images for training, the network would become able to detect pneumonia and normal classes very well, but not the COVID-19 class because of the unbalanced dataset. This condition is not our goal because the main purpose here is to achieve good results in detecting COVID-19 cases and not to identify wrong COVID-19 cases.

The best way to solve this problem is to make the dataset balanced and provide the network almost equal data of each class when training, so that the network will learn to identify all classes. Here

because we do not access more open-source datasets of COVID-19 to increase this class data, we chose the number of pneumonia and normal classes almost equal to the COVID-19 number of images.

We decided to train the networks for 8 consecutive phases. In each phase, we selected 250 cases of normal class and 234 cases of pneumonia class along with the 149 COVID-19 cases. In total, we had 633 cases for each training phase. All of the COVID-19 images and 34 of the pneumonia images were common between each training phase and 250 normal cases, and 200 pneumonia cases were unique in each training phase. The common 149 COVID-19 and 34 pneumonia cases between all the training phases were from the covid chestxray dataset; and the rest of the data were from the other dataset. Based on this categorizing, our training set includes 8 phases and 3783 images.

By doing so, the network sees an almost equal number of images for each class, so it helps to improve the COVID-19 detection along with detecting pneumonia and normal cases. But as we had more pneumonia and normal cases, we showed the network different pneumonia and normal cases with COVID-19 cases in each phase. Implementing this method results in two advantages. One is that the network learns COVID-19 class features better along with the other classes; second, the normal and pneumonia classes' detection improves greatly.

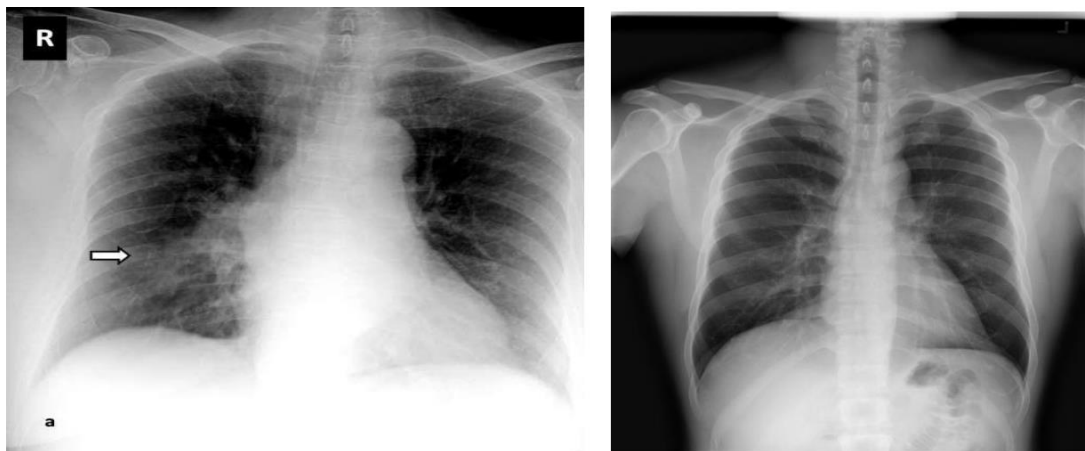


Fig 5.1 Sample images of the dataset

5.3 TRAINING PHASE

We described in the dataset that we allocated 8 phases for training. For reporting more reliable results, we chose five folds for training, where in every fold the training set was made of 8 phases as it is mentioned. We have trained ResNet50V2; Xception; and a concatenation of Xception and ResNet50V2 neural networks based on the explained method. This concatenated Neural Network has shown higher accuracy compared to others. As we have tested several networks in our project, the Xception and ResNet50V2 networks work as well or better than others in extracting deep features. By concatenating the output features of both networks, we helped the network learn to classify the input image from both feature vectors, which resulted in better accuracy.

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Training Phase	100	100	100
Horizontal/Vertical flipping	Yes	Yes	Yes
Zoom Range	5%	5%	5%
Rotation Range	0–360°	0–360°	0–360°
Width/Height	5%	5%	5%
Shift Range	5%	5%	5%
Re-scaling	1/255	1/255	1/255

Based on table trained the networks using the Categorical cross-entropy loss function and Nadam optimizer. The learning rate was set to 1e-4. Trained the network for 100 epochs in each training phase and, because of having 8 training phases, the models were trained for 800 epochs.

CHAPTER 6

TESTING

6.1 INTRODUCTION

Validated our networks on 31 cases of COVID-19, 4420 cases of pneumonia, and 6851 normal cases. The reason our training data was less than the validation data is that we had a few cases of COVID-19 among many normal and pneumonia cases. Therefore, we could not use many images from the two other classes with COVID-19 fewer cases for training, because it would have made the network not learn COVID-19 features. To solve this issue, we selected 3783 images for training in 8 different phases. Evaluated our network on the remainder of the data so that our trained network's ultimate performance would be clear. It must be noticed that exceptionally, in fold3, we had 30 cases of COVID-19 for validation, and 150 other cases were allocated for training

6.2 TEST RESULTS AND REPORTS.

It is noteworthy that we used transfer learning in the training process. For all of the networks, we used the pre-trained ImageNet weights at the beginning of the training and then resumed training based on the explained conditions on our dataset. We also used the accuracy metric for monitoring the network results on the validation set after each epoch to find the best and most converged version of the trained network. Reported the four different metrics for evaluating our network for each of the three class as follows:

$$\text{Accuracy(for each class)} = \frac{TP+TN}{TP+FP+TN+FN}$$

$$\text{Specificity} = \frac{TN}{TN+FP}$$

$$\text{Sensitivity} = \frac{TP}{TP+FN}$$

$$\text{Precision} = \frac{TP}{TP+FP}$$

The number of true and false positive and false negative for each class

Fold	Network	COVID-19 Correct detected	COVID-19 Not detected	COVID-19 Wrong detected	PNEUMONIA Correct detected	PNEUMONIA Not detected	PNEUMONIA Wrong detected	NORMAL Correct Detected	NORMAL Not detected	NORMAL Wrong detected
1	Xception	26	5	101	3983	437	569	6245	606	378
	ResNer50V2	27	4	96	3858	562	480	6334	517	507
	Concatenated	26	5	68	3745	675	309	6526	325	628
2	Xception	23	8	42	3874	546	409	6426	425	528
	ResNer50V2	22	9	67	3659	761	501	6340	511	713
	Concatenated	23	8	27	3913	507	434	6413	438	492
3	Xception	21	9	28	3942	478	436	6411	440	463
	ResNer50V2	22	8	97	3770	650	392	6433	418	587
	Concatenated	25	5	35	3847	573	342	6502	349	550
4	Xception	22	9	42	3818	602	433	6411	440	576
	ResNer50V2	22	9	78	4015	405	758	6065	786	364
	Concatenated	26	5	77	3860	560	480	6340	511	519
5	Xception	21	10	41	4041	379	502	6335	516	362
	ResNer50V2	21	10	42	3604	816	284	6549	302	802
	Concatenated	24	7	43	3941	479	390	6442	409	462

Some of the evaluation metrics have been reported in this table.

Fold	Network	Accuracy	Covid-19 Sensitivity	Pneumonia Sensitivity	Normal Sensitivity
1	Xception	90.72	83.87	90.11	91.15
	ResNet50V2	90.41	87.09	87.28	92.45
	Concatenated	91.10	83.87	84.72	95.25
	Xception	91.33	74.19	87.64	93.79
2	ResNet50V2	88.66	70.96	82.78	92.54
	Concatenated	91.56	74.19	88.52	93.60
	Xception	91.79	70	89.18	93.57
3	ResNet50V2	90.47	73.33	85.29	93.89
	Concatenated	91.79	83.33	87.03	94.90
	Xception	90.70	70.96	86.38	93.57
4	ResNet50V2	89.38	70.96	90.83	88.52

5	Concatenated	90.47	83.87	87.33	92.54
	Xception	91.99	67.74	91.42	92.46
	ResNet50V2	90.01	67.74	81.53	95.59
A ve ra ge	Concatenated	92.08	77.41	89.16	94.03
	Xception	91.31	73.35	88.95	92.91
	ResNet50V2	89.79	74.02	85.54	92.60
	Concatenated	91.40	80.53	87.35	94.06

Fold	Network	Accuracy	Covid-19 Precision	Pneumonia Precision	Normal Precision
1	Xception	90.72	20.47	87.50	94.29
	ResNet50V2	90.41	21.95	88.93	92.58
	Concatenated	91.10	27.65	92.37	91.22
	Xception	91.33	35.38	90.45	92.40
2	ResNet50V2	88.66	24.71	87.95	89.89
	Concatenated	91.56	46	90.01	92.87
	Xception	91.79	42.85	90.04	93.26
3	ResNet50V2	90.47	18.48	90.58	91.63
	Concatenated	91.79	41.66	91.83	92.20
	Xception	90.70	34.37	89.81	91.75
4	ResNet50V2	89.38	22	84.11	94.33
	Concatenated	90.47	25.24	88.94	92.43
	Xception	91.99	33.87	88.95	94.59

5	ResNet50V2	90.01	33.33	92.69	89.08
	Concatenated	92.08	35.82	90.99	93.30
	Xception	91.31	33.39	89.35	93.26
Average	ResNet50V2	89.79	24.09	88.85	91.50
	Concatenated	91.40	35.27	90.83	92.40

$$\text{Accuracy (for all the classes)} = \frac{\text{Number of Correct Classified Images}}{\text{Number of All Images}}$$

In these equations, *TP* (True Positive) is the number of correctly classified images of a class, *FP* (False Positive) is the number of the wrong classified images of a class, *FN* (False Negative) is the number of images of a class that have been detected as another class, and *TN* (True Negative) is the number of images that do not belong to a class and were not classified as belonging to that class.

CHAPTER 7

RESULTS

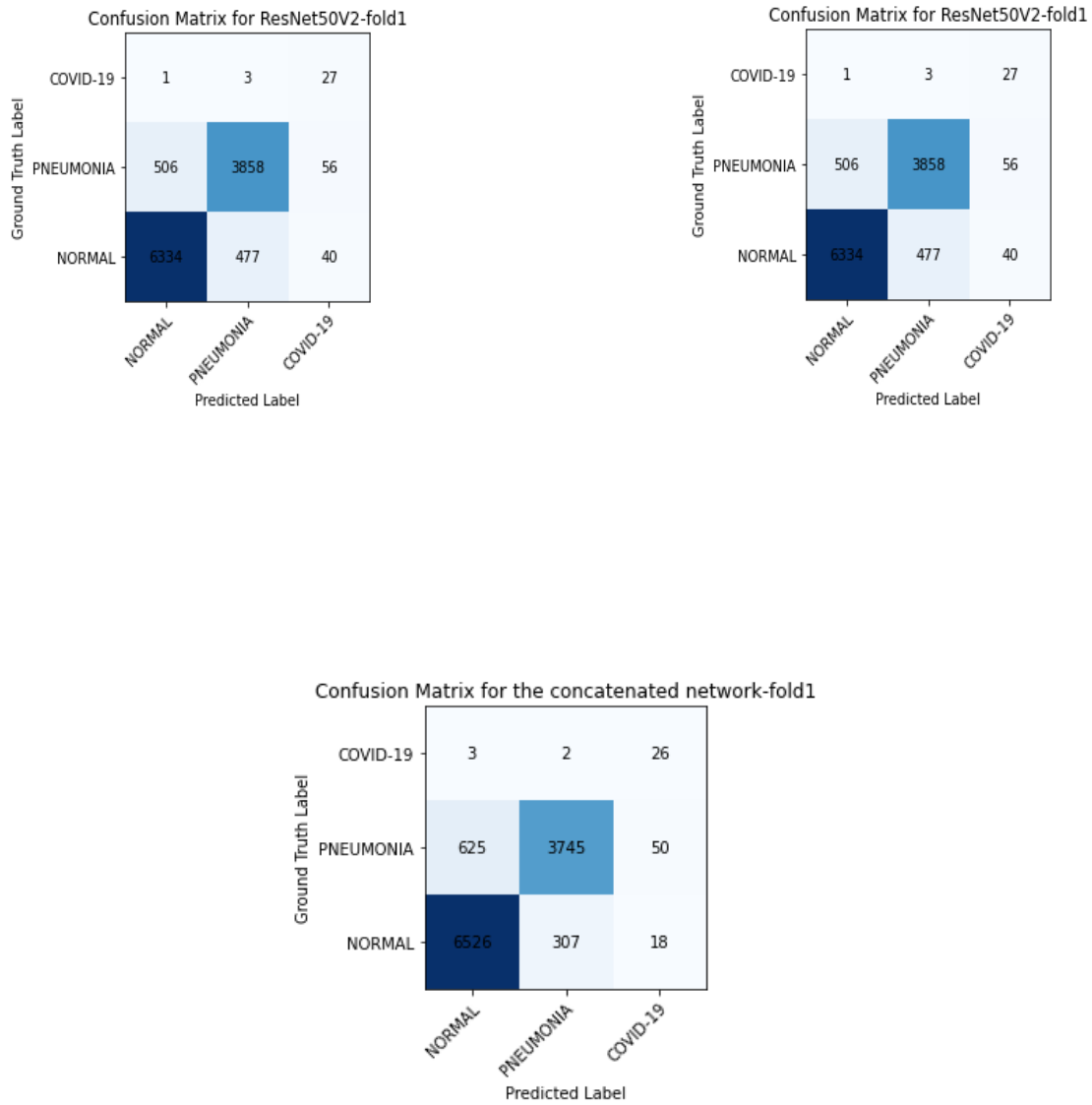


Fig 7.1 Output

CHAPTER 8

CONCLUSION AND FUTURE WORK

In this project, presented a concatenated neural network based on Xception and ResNet50V2 networks for classifying the chest X-ray images into three categories of normal, pneumonia, and COVID-19. We used two open-source datasets that contained 180 and 6054 images from patients infected with COVID-19 and pneumonia, respectively, and 8851 images from normal people. As we had a few images of the COVID-19 class, we proposed a method for training the neural network when the dataset is unbalanced. We separated the training set into 8 successive phases, in which there were 633 images (149 COVID-19, 234 pneumonia, 250 normal) in each phase.

Our training set included 3783 images, and the rest of the images were allocated for evaluating the network. We tried to test our model on a large number of images so that our real achieved accuracy would be clear. We achieved an average accuracy of 99.50%, and 80.53% sensitivity for the COVID-19 class, and an overall accuracy equal to 91.4% between five folds. We hope that our trained network that is publicly available will be helpful for medical diagnosis. We also hope that in the future, larger datasets from COVID-19 patients become available, and by using them, the accuracy of our proposed network increases further.

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